Restaurant Management System (SwiftBite)

System Architecture Specification & Technical Documentation — Category: Full Stack E-Commerce & Hospitality

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Repository: https://github.com/rakshapatil526/Restaurant-Management-System

Status: Verified Repository

Doc ID: SPEC-RESTAURANT-MANAGEMENT

Date: March 2026

1. Executive Summary & Problem Statement

SwiftBite is a full-stack food delivery and restaurant operations platform engineered with React.js and Django REST Framework. The application provides an intuitive consumer-facing food discovery and checkout experience coupled with a comprehensive administrative back-office for real-time menu management, category classification, and order tracking.

The system solves common restaurant operational challenges including out-of-sync menus, manual order logging, and lack of real-time sales visibility.

2. Core System Features & Modules

• **Authentication Module:** Customer signup, JWT-secured authentication, password visibility toggles, and one-click demo login presets.

• **Interactive Menu Catalog:** Categorized navigation (Pizza, Burgers, Biryani, Chinese, Desserts, Drinks) with Veg/Non-Veg indicators.

• **Cart & Checkout Pipeline:** Interactive item steppers, subtotal calculation, delivery charges, Cash on Delivery (COD) and Online payment mock options.

• **Admin Kitchen Operations:** Real-time order state machine (Placed → Preparing → Out for Delivery → Delivered) and menu CRUD console.

3. Technology Stack

Tier	Technology	Implementation Purpose
Frontend UI	React 18, Vite, Tailwind CSS	Component-driven user interface with responsive product grids and cart drawers
Backend API	Django 5, DRF, Python 3	RESTful endpoints for catalog filtering, cart persistence, and order placement
Database	PostgreSQL / SQLite	Relational storage for categories, menu items, orders, and order items
State & Network	Axios, Context API	Persistent cart management, auth token interceptors, dynamic error handling

4. Database Architecture & Relationships

Model	Primary Key	Key Attributes & Relationships
Category	id (AutoField)	name (unique), slug, image_url, is_active
Product	id (AutoField)	category_id (FK: Category), name, price, prep_time, is_veg, rating, image
Order	id (AutoField)	user_id (FK: User), order_number, total_amount, payment_mode, order_status
OrderItem	id (AutoField)	order_id (FK: Order), product_id (FK: Product), quantity, unit_price

5. API Architecture & Limitations

• **Endpoints:** `/api/products/`, `/api/products/?category={id}`, `/api/orders/`, `/api/orders/{id}/status/`, `/api/auth/token/`.

• **Security:** JWT token authentication; order placement scoped to active user credentials.

• **Limitations & Roadmap:** Integrated payment gateway uses simulated confirmation callback; future enhancements include WebSocket order status push notifications.